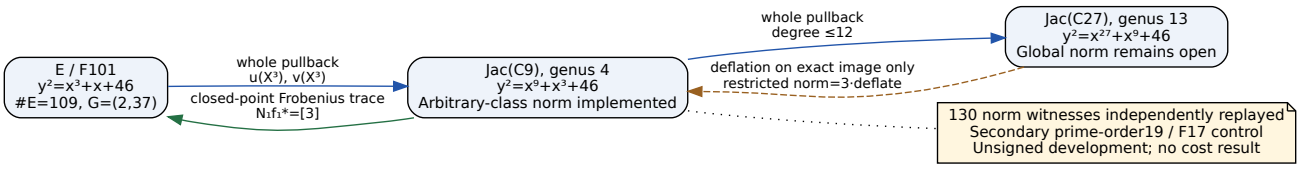
Cubic covers: the arbitrary-class norm prerequisite

2026-11-11 · Chained ECDLP Transfers · crypto PR3877 / queue3904

Standing: unsigned and unverified. This implements a bounded missing norm operation on the existing prime-field cover family. It claims no new cover, solver speedup or cross-characteristic descent. [pcc:unverified]



Exact operation and proof boundary

On E/F101:$y^2=x^3+x+46$, the full rational group has prime order109, with G=(2,37). The cubic covers have genera4 and13. For canonical (u,v), whole pullback substitutes X^3 in both polynomials; degree≤3g fits the target genus3g+1. Coefficient deflation is a checked inverse on the exact image. It is not a global genus13 norm. [pcc:unverified]

The actual genus4 norm factors every support polynomial of degree≤4 and traces the projected elliptic point in Fp[X]/q over all Frobenius conjugates, with support multiplicity. Factor reconstruction, orbit closure and descent are checked. All canonical genus4 inputs are within the algorithm's domain; the test set is130 fixed translated classes, not exhaustive enumeration of that Jacobian. The prototype restricts checked primes to5..101. [pcc:unverified]

Norm-after-first-pullback is[3]. The restricted two-hop composition gives[9], invertible by97 modulo109; first recovery uses73. These recover source points, not unknown scalars. Four Lean lemmas establish coefficient left-inverse/injectivity and closed-orbit trace invariance. The12-declaration gate passes. Finite-field and Picard geometry remain written arguments. PCC-e1a827 remains unsigned. [pcc:unverified]

Correctness evidence

All109 source scalars pass both pullbacks and first norms. A predeclared E/F17:$y^2=x^3+2x+2$,order19 control checks361 ordered subgroup pairs and ramification, reaching degree12 representatives on genus13. The fixed130 norm witnesses include irreducible support degrees3 and4. [pcc:unverified]

The independent root replay imports no new norm code. Existing genus4/13 arithmetic reconstructs all classes, existing affine elliptic arithmetic reconstructs all expected norms, and a distinct Frobenius-gcd criterion confirms cubic/quartic support. Receipt cubic-independent-01.txt and witness TSV are bound by PCC-2d5c08. Both Jacobian engines use Cantor; producer-authored agreement is not a signed independent verdict. [pcc:unverified]

Restrictions, retained negatives and next task

What the quadratic obstruction does establish

Odd full elliptic group order implies an irreducible Weierstrass cubic. If α is a root of $f(X^2+s)$, α^2+s has degree3, so the tower law makes the degree of α divisible by3. Thus there are no rational linear or quadratic factors. The existing three-individually-rational-factor Richelot constructor and rational-Weierstrass chart cannot start here. This is checked directly on the E109 sextic. [pcc:unverified]

An unordered triple of nonrational quadratic factors can nevertheless be Frobenius-stable. No universal Richelot obstruction follows. Next: implement balanced genus2 arithmetic with two infinities, validate it against the existing odd-chart engine, then assess explicit non-split kernel descent. This remains prospective work. [pcc:unverified]

Prior art and unresolved computational cost

The existing Enge20261024 study already constructs the C(d,s) family, analyzes its published generic bound and tests rational-point collectors. Its open problem prime-cover-projected-relation-cost-missing asks for closed primes, norm, replay and total-cost evidence. This work addresses a bounded arithmetic prerequisite. The original relation-yield and linear-algebra gaps remain. An upper-bound comparison is not a lower bound excluding specialized methods.

Increasing genus alone gives no advantage on the embedded order109 subgroup. The image is recognizable by coefficient deflation. No relation collector, DLP solver, benchmark, cryptographic-size field or prime-field descent has been demonstrated. [pcc:unverified]

Integrity and reproducibility

Producer: cryptanalysis/experiments/chained-transfer-reference/src/cubic_transfer.rs. Independent replay: crypto/examples/chained_cubic_norm_reference.rs. Run the consumer check.sh with Rust1.99.1 on PATH; run cargo test –release –example chained_cubic_norm_reference from crypto. Core Lean4.35.0 audits lean/CubicPullback.lean using tools/lean/proof-gate.sh. The checkpoint manifest binds the producer revision and witnesses. First compile/PATH failures remain retained separately from subsequent passes. [pcc:unverified]

The full crypto library result remains6184 pass/13 fail/133 ignored. Restoring tracked fixtures fixes6 failing tests;5 still need archived raw data and2 need unavailable kernel clock metadata. The failed full run remains failed. PCC-8ffeb3 preserves it. Crypto’s frozen ledger fork blocks merging PR3877. No history, test or evidence was rewritten. [pcc:unverified]

S3 admission and live IsoLab budget/capacity are unavailable, so no experiment or paid job was launched. KB access is absent and crypto-autoresearcher’s vault remains locked; its integration handoff stays in crypto.

Sources: Galbraith corrected ch.8 §8.3 and ch.10 §§10.3–10.4; research/enge_primefield_cover_20261024/REPORT.md and its native source; problems/prime-cover-projected-relation-cost-missing.md. Full source-linked account: CUBIC_NORM_20261111.md. Canonical transfer/leaderboard graphs remain unchanged because the evidence is unsigned and contains no performance edge.